LAB EXPERIMENT RECORD

SET 22: TIME SERIES ANALYSIS

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	15 OCTOBER 2026
Day	Thursday

Aim

To analyze monthly sales data using a time series line chart, a scatter plot, an interactive dashboard, and an autocorrelation plot in R and Shiny

Dataset: Monthly Sales

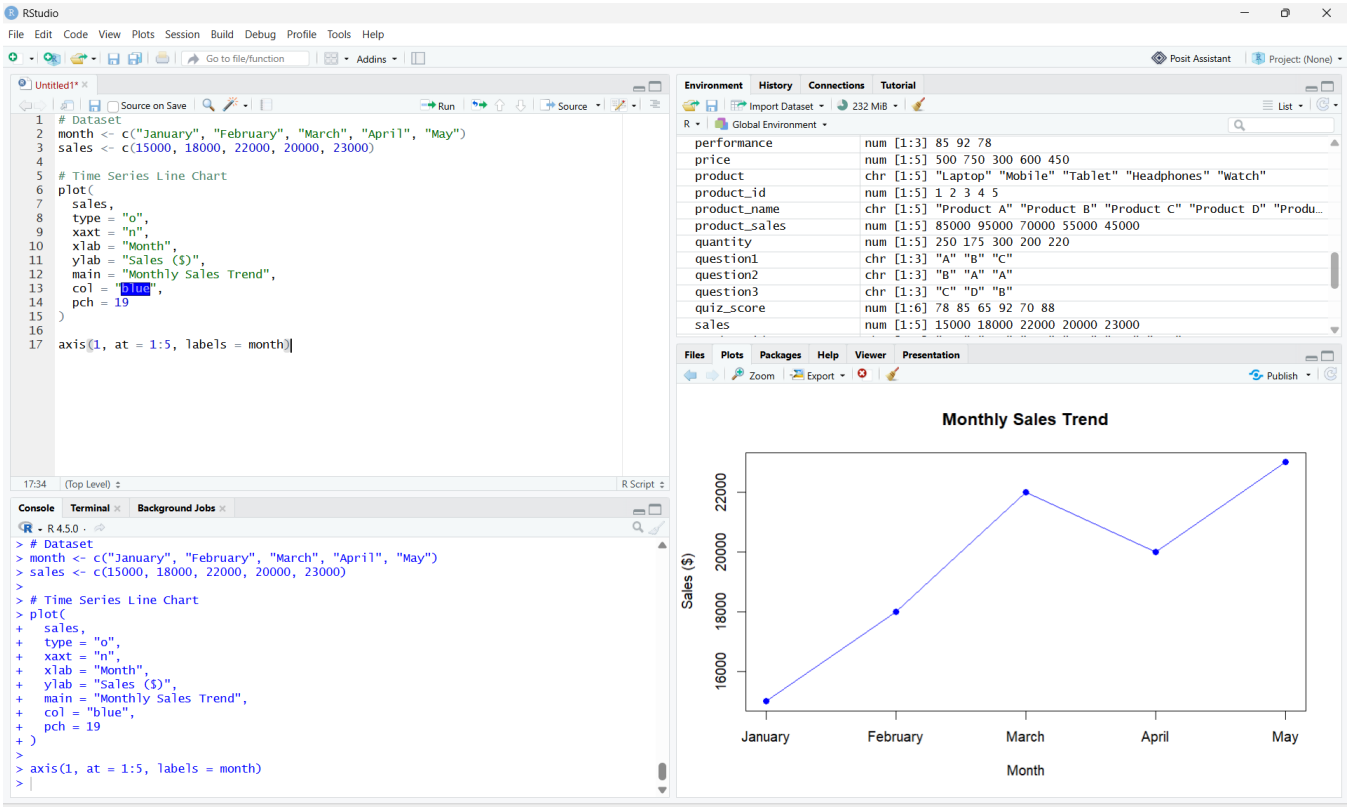
Month	Sales ($)
January	15000
February	18000
March	22000
April	20000
May	23000

PROGRAM 1: Time Series Line Chart of Monthly Sales

To visualize the trend in monthly sales using a time series line chart with appropriate axis labels and title.

```
# # Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)
# Time Series Line Chart
plot(
  sales,
  type = "o",
  xaxt = "n",
  xlab = "Month",
  ylab = "Sales ($)",
  main = "Monthly Sales Trend",
  col = "blue",
```

```
  pch = 19
)
axis(1, at = 1:5, labels = month)
```



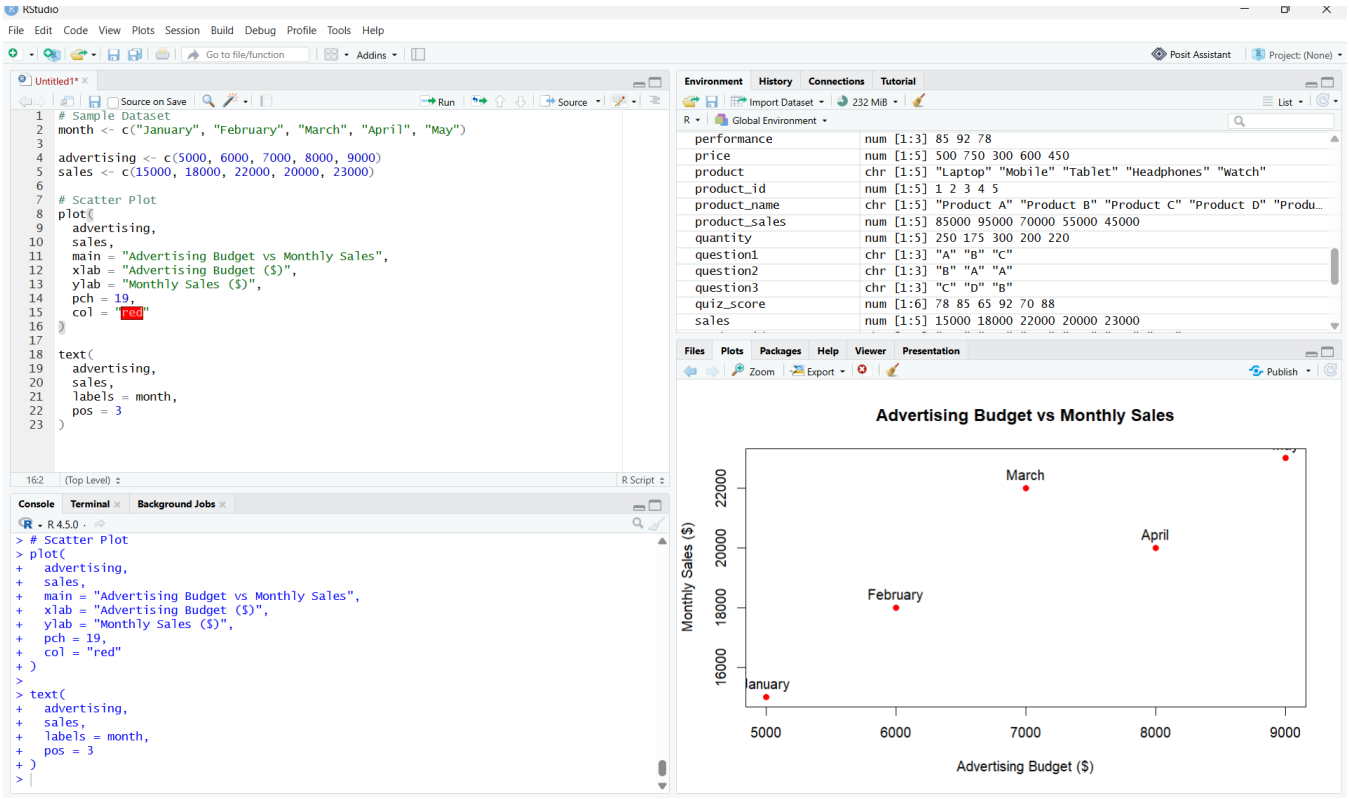
Observation: Sales increase from January to March, decrease slightly in April, and reach their highest value of $23,000 in May.

PROGRAM 2: Scatter Plot of Advertising Budget vs Monthly Sales

To explore the relationship between advertising expenditure and monthly sales using a scatter plot

```
# Sample Dataset
month <- c("January", "February", "March", "April", "May")
advertising <- c(5000, 6000, 7000, 8000, 9000)
sales <- c(15000, 18000, 22000, 20000, 23000)
# Scatter Plot
plot(
  advertising,
  sales,
  main = "Advertising Budget vs Monthly Sales",
  xlab = "Advertising Budget ($)",
  ylab = "Monthly Sales ($)",
  pch = 19,
  col = "red"
)

text(
  advertising,
  sales,
  labels = month,
  pos = 3
))
```



Observation: the sample values show a generally positive relationship between advertising budget and sales. However, this does not prove that increased advertising causes higher sales..

PROGRAM 3: Interactive Monthly Sales Dashboard

To create an interactive dashboard combining a line chart for monthly sales and a pie chart displaying sales distribution across products.

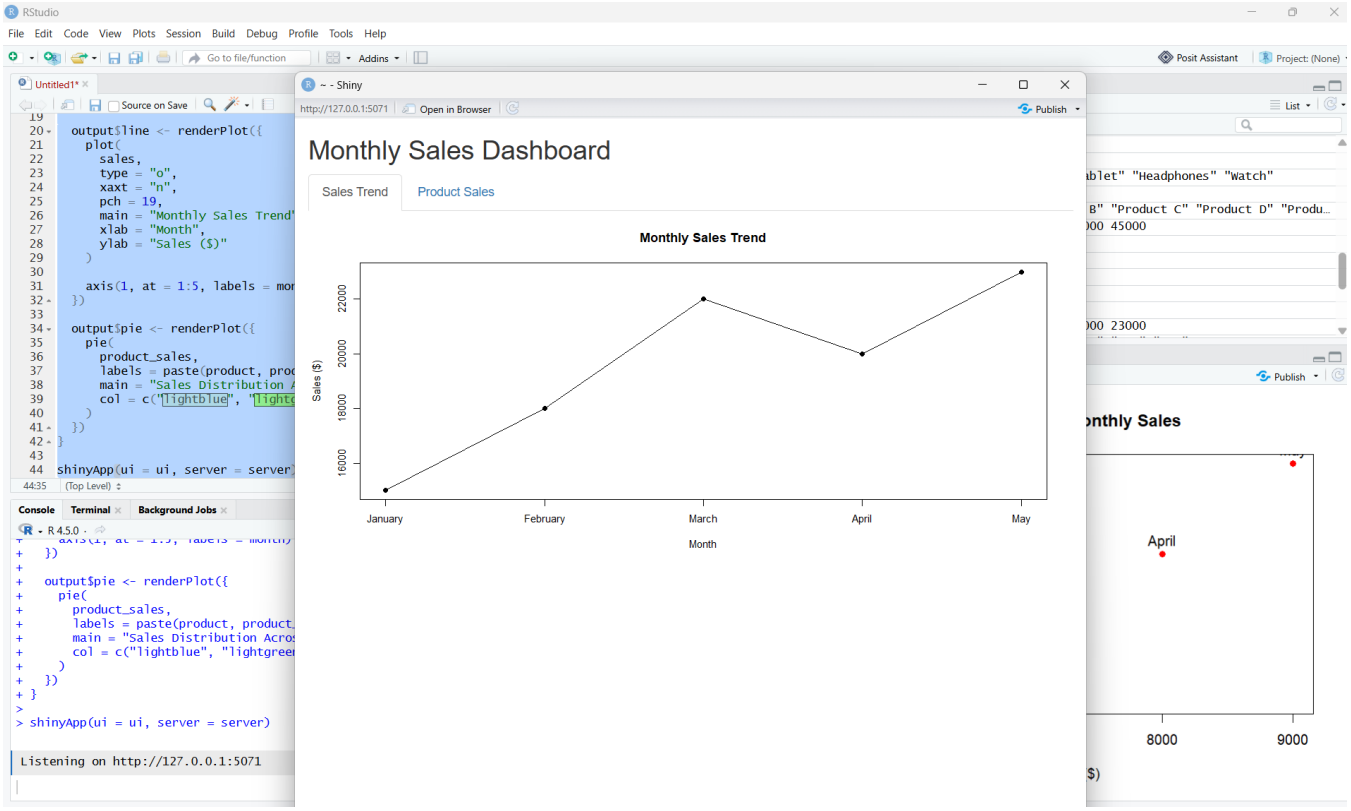
```
library(shiny)


month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)


product <- c("Product A", "Product B", "Product C")
product_sales <- c(30000, 25000, 21000)


ui <- fluidPage(
  titlePanel("Monthly Sales Dashboard"),


  tabsetPanel(
    tabPanel("Sales Trend", plotOutput("line")),
    tabPanel("Product Sales", plotOutput("pie"))
  )
)
server <- function(input, output) {
    axis(1, at = 1:5, labels = month)
  })
  output$pie <- renderPlot({
    pie(
      product_sales,
      labels = paste(product, product_sales),
      main = "Sales Distribution Across Products",
      col = c("lightblue", "lightgreen", "orange")
    )
  })
}
shinyApp(ui = ui, server = server))
```



Insight: The dashboard allows users to switch between the monthly sales trend and the illustrative product sales distribution..

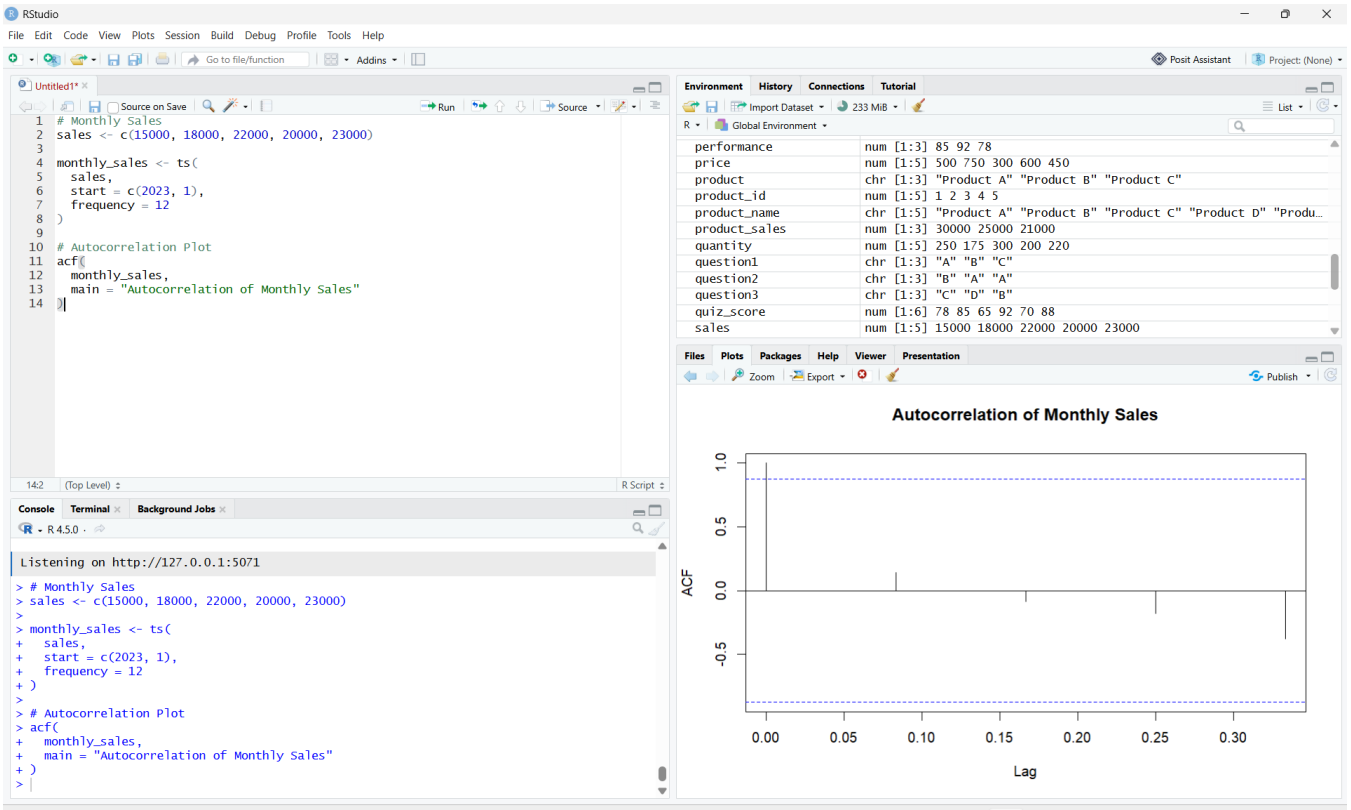
PROGRAM 4: Autocorrelation Plot

To create an autocorrelation plot to examine possible patterns and seasonality in monthly sales.

```
# Monthly Sales
sales <- c(15000, 18000, 22000, 20000, 23000)


monthly_sales <- ts(
  sales,
  start = c(2023, 1),
  frequency = 12
)


# Autocorrelation Plot
acf(
  monthly_sales,
  main = "Autocorrelation of Monthly Sales"
)}
```



Observation: The autocorrelation plot shows correlations at different time lags. Since only five monthly observations are available, reliable conclusions about seasonality cannot be made.

Procedure

- Create vectors for months and sales values.
- Use plot() and axis() to visualize the monthly sales trend.
- Create advertising and sales vectors and use plot() to examine their relationship.
- Use Shiny to combine the sales line chart and product sales pie chart in an interactive dashboard.
- Convert the sales values into a time series using ts() and create an autocorrelation plot using acf().
- Run each program in RStudio and observe the results.

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Result

Monthly sales data was successfully analyzed using a line chart, scatter plot, interactive dashboard, and autocorrelation plot. The visualizations helped examine sales trends and possible relationships in the data.